```
import matplotlib.pyplot as plt
import numpy as np
from scipy import stats

# ========================================
# 1. DATA FROM LAB SHEET & PARTNER DATA
# ========================================
# Infill densities (%)
infill = np.array([10, 50, 100])

# PLA Data (Shore D) - From Yihan NTU image
# Format: [Trial 1, Trial 2, Trial 3]
pla_trials = np.array(
    [
        [55.5, 58.5, 56.0],  # 10%
        [67.5, 72.5, 70.0],  # 50%
        [78.5, 81.5, 80.0],  # 100%
    ]
)

# TPU Data (Shore A) - From handwritten table image
tpu_trials = np.array(
    [
        [74.5, 81.5, 80.0],  # 10%
        [89.5, 88.5, 91.5],  # 50%
```

```
        [93.0, 93.0, 92.0],  # 100%
    ]
)

# Calculate Means & Standard Deviations
pla_means = np.mean(pla_trials, axis=1)  # [56.67, 70.00, 80.00]
pla_stds = np.std(pla_trials, axis=1, ddof=1)

tpu_means = np.mean(tpu_trials, axis=1)  # [78.67, 89.83, 92.67]
tpu_stds = np.std(tpu_trials, axis=1, ddof=1)

# Raw data arrays for scatter points
infill_flat = np.repeat(infill, 3)
pla_flat = pla_trials.flatten()
tpu_flat = tpu_trials.flatten()

# ========================================
# 2. LINEAR REGRESSION ANALYSIS
# ========================================
# PLA Linear Regression
slope_pla, intercept_pla, r_pla, p_pla, _ = stats.linregress(infill, pla_means)
r2_pla = r_pla**2

# TPU Linear Regression
slope_tpu, intercept_tpu, r_tpu, p_tpu, _ = stats.linregress(infill, tpu_means)
```

```
r2_tpu = r_tpu**2


# Print values to terminal
print("=== PLA (Shore D) Regression ===")
print(
    f"Equation: y = {slope_pla:.4f}x + {intercept_pla:.4f} | R² = {r2_pla:.4f}"
)


print("\n=== TPU (Shore A) Regression ===")
print(
    f"Equation: y = {slope_tpu:.4f}x + {intercept_tpu:.4f} | R² = {r2_tpu:.4f}"
)


# ========================================
# 3. PLOTTING
# ========================================
x_trend = np.linspace(5, 105, 100)


plt.figure(figsize=(9, 6), dpi=150)


# --- PLA Plotting (Red theme) ---
plt.scatter(
    infill_flat,
    pla_flat,
    color="#d9534f",
```

```
    alpha=0.4,
    marker="o",
    label="PLA Raw Trials",
)
plt.errorbar(
    infill,
    pla_means,
    yerr=pla_stds,
    fmt="o",
    color="#c9302c",
    ecolor="black",
    capsize=4,
    markersize=8,
    label="PLA Mean Hardness",
)
plt.plot(
    x_trend,
    slope_pla * x_trend + intercept_pla,,
    color="#c9302c",
    linestyle="--",
    linewidth=2,
    label=f"PLA Fit: y = {slope_pla:.2f}x + {intercept_pla:.2f} ($R^2$ = {r2_pla:.3f})",
)

# --- TPU Plotting (Blue theme) ---
```

```
plt.scatter(
    infill_flat,
    tpu_flat,
    color="#0275d8",
    alpha=0.4,
    marker="s",
    label="TPU Raw Trials",
)
plt.errorbar(
    infill,
    tpu_means,
    yerr=tpu_stds,
    fmt="s",
    color="#025aa5",
    ecolor="black",
    capsize=4,
    markersize=8,
    label="TPU Mean Hardness",
)
plt.plot(
    x_trend,
    slope_tpu * x_trend + intercept_tpu,
    color="#025aa5",
    linestyle="-.",
    linewidth=2,
```

```
    label=f"TPU Fit: y = {slope_tpu:.2f}x + {intercept_tpu:.2f} ($R^2$ = {r2_tpu:.3f})",
)

# Formatting
plt.title(
    "Lab 2: Infill Density vs. Shore Hardness (PLA & TPU)",
    fontsize=13,
    fontweight="bold",
    pad=12,
)
plt.xlabel("Infill Density (%)", fontsize=11)
plt.ylabel(
    "Shore Hardness\n(PLA: Shore D | TPU: Shore A)", fontsize=11
)
plt.xlim(0, 105)
plt.ylim(40, 100)
plt.grid(True, linestyle=":", alpha=0.6)
plt.legend(loc="lower right", fontsize=9, framealpha=0.95)

plt.tight_layout()
plt.savefig("lab2_regression_plot.png", dpi=300)
plt.show()
```